

Novelty is a novel approach to modelling selection criteria in evolutionary algorithms and has been proven as viable technique of avoiding pitfalls of false optima in tasks abundant with them, such as solving mazes. Rather than closing the topic however, this finding opened other problems to explore: How to properly apply novelty in tasks that yield slightly better to conventional approaches? How to properly model behavioural space necessary for novelty computation? In this thesis we investigate use of novelty in selected reinforcement learning tasks, it's combinations with classical fitness and propose behaviour space models for the respective RL tasks.